

CONFIDENTIAL INVESTOR & TECHNICAL BRIEF
H~M Trading Institute

Quant Research OS — AI-Assisted Trading Research for African Retail Investors
Subscription SaaS, Nigerian Entity, Production-Ready Platform

Prepared: August 2026 (Updated August 23, 2026)
Status: MVP Live + Quant Research OS Roadmap (Phases 0–10)
Contact: support@hmtrade.business | hmtrade.business

EXECUTIVE SUMMARY

H~M Trading Institute is a multi-user subscription platform delivering institutional-grade AI trading research to retail investors in Nigeria and across Africa. Users bring their own DeepSeek API key (BYOK model); the platform provides signup, secure key storage (Vault encryption), run queueing, a 462-alpha research engine, swarm orchestration, live progress streaming, artifact rendering, and NGN billing via Paystack.

The MVP is live and verified end-to-end: prompt → queued → worker claim → engine execution → artifacts → completed. The platform is architected as a strangler-fig migration toward a Quant Research OS — a 10-phase upgrade that unifies AI research agents, canonical market data, strategy development, ML/RL experimentation, realistic backtesting, paper trading, risk controls, and controlled live execution.

What makes this unique: BYOK economics mean zero LLM margin risk and true unlimited runs per tier. The infrastructure is capital-efficient (Supabase-only MVP at ~\$100/mo). Security is database-level (RLS, SECURITY DEFINER RPCs, Vault encryption). The team is solo but disciplined: 74 hermetic worker tests, production monitoring (Sentry), structured JSON logs, and a constitution-driven roadmap that never breaks the live loop.

1. THE PROBLEM

Retail investors in Nigeria and across emerging markets face a research gap:

- Professional-grade trading research (alpha screening, backtesting, synthesis) is locked inside institutions or priced for Western salaries.
- Local tools (fractional brokerage, savings apps) are execution platforms — they let users buy but not think.
- Fragmented tools assume US/EU infrastructure (USD pricing, foreign card rails).
- At exactly the moment frontier LLMs made research cheap, there is no accessible, localized research layer for African retail traders.

Two structural facts make this solvable now: (1) frontier LLMs (DeepSeek) are available per-token to anyone with a card, (2) BYOK lets a small company offer unlimited AI research without bearing model cost — the customer pays for intelligence, the platform sells the product around it.

2. THE SOLUTION

2.1 Product (MVP — Live Now)

A web application where users sign up, connect their own DeepSeek key, and launch AI trading research agents. Natural-language prompts become runs that scan markets, backtest strategies, and return rendered artifacts (charts, tables, reports) with live progress streaming.

Subscription Tiers (NGN, unlimited runs per tier):

Tier	Price (NGN/mo)	≈ USD	Capabilities
Starter	₦20,000	~\$12	Single-agent research runs
Pro	₦35,000	~\$22	Starter + Swarm (30 presets)
Premium	₦75,000	~\$47	Pro + Attachments (CSV/XLSX/JSON)

2.2 Why BYOK (Bring Your Own Key)

- Zero LLM margin risk: platform never proxies or marks up model calls.
- Trust: users' keys encrypted at rest in Supabase Vault; platform never sees or spends them.

- Unlimited positioning: since tokens are customer's, 'unlimited runs' is a true statement.
- Simplified compliance: no handling of customer LLM content through our payment flow.
- Defensible unit economics: 90%+ gross margin by construction (customer pays LLM provider directly).
- Multi-provider: DeepSeek (cloud, sk-... API key) OR Ollama (self-hosted base URL). Worker resolves provider per-user at run time (DeepSeek-first, Ollama fallback).

3. ARCHITECTURE & INFRASTRUCTURE

3.1 Live Stack (MVP — Today)

Frontend: React 19 + Vite + TypeScript (Tradi/frontend/), deployed on Vercel (hmtrade.business).

Backend: Supabase-only — PostgreSQL 17, Auth (JWT), Storage (artifacts), Vault (BYOK encryption), Realtime, Edge Functions.

Worker: Python polling loop (vibe-trading-saas/worker/), Railway container, subprocess-per-run isolation.

Engine: Vibe-Trading vendored (MIT, HKUDS/upstream), 462 alphas, 8 backtest engines, ReAct agent, 30 swarm presets.

Billing: Paystack NGN (test mode live, can switch to live with one click).

Security: RLS on all user-facing tables, SECURITY DEFINER RPCs for privileged operations, Vault encryption for keys, FOR UPDATE SKIP LOCKED claim for safe multi-worker execution.

Observability: Sentry (frontend + worker), structured JSON logs with run IDs, /health endpoint (Railway health checks).

3.2 Target Infrastructure (Quant Research OS — Phases 1–10)

Cloudflare (edge, control plane): DNS/WAF, Workers gateway, R2 (data lake), Queues.

Vercel: SPA (User App + Admin App routes).

Railway container (same, two processes):

- FastAPI modular monolith (platform/, port 8000) — research-governance API, experiments, hypotheses, datasets, promotion.
- hm-worker (existing, port 9100) — run polling, engine execution, provenance

capture.

Supabase (transactional source of truth): PostgreSQL 17, Auth, Vault, Storage, RLS, Realtime.

Upstash Redis (free tier): health probes, queue skeleton (fail-open, never blocks the loop).

This topology never breaks the live run loop (strangler-fig: new stands beside old; D16 invariant).

4. DATA & SECURITY MODEL

4.1 Schema (Canonical, PostgreSQL 17)

Live tables (MVP):

- profiles — public profile (id = auth.uid()).
- plans — tiers (starter/pro/premium), NGN price, Paystack plan code.
- subscriptions — active plan, provider subscription ID, period, status.
- agent_runs — run queue + state machine (queued/running/completed/failed/timeout/cancelled).
- agent_artifacts — result files (storage_path, kind).
- user_api_keys — BYOK metadata (last-4, configured_at); secret in Vault (zero-policy RLS).
- webhook_events — idempotency ledger (provider_event_id UNIQUE, Paystack re-verified).
- admin_users, audit_logs — admin gate + audit trail (append-only, zero-policy RLS).

Phase 1 additions:

- projects — tenant-scoped, auto-default per user (trigger + backfill).
- hypotheses — falsifiable-criterion statements, status enum.
- experiments — provenance record (dataset/feature/strategy/model version, param set, windows, seed, commit, data hash, status lifecycle, result JSONB).
- dataset_registry — read-mostly, Phase 2 populates (LSE OHLCV as first source).
- promotion_events — audited status transitions (service-role only, via record_experiment_status RPC).

4.2 Security Posture

Row isolation: RLS on every user-facing table (`user_id = auth.uid()`).

Privileged data: zero policies (default deny), reachable only through gated RPCs (SECURITY DEFINER, service-role only).

BYOK keys: Supabase Vault (encrypted at rest, no plaintext path); worker decrypts server-side via `worker_get_user_api_key` RPC.

JWT boundary: Supabase JWTs (JWKS-verified, exp-checked); FastAPI validates on every request; tenant ID = `jwt.sub`.

Webhooks: Paystack HMAC-SHA512 constant-time verification + API re-verify; `webhook_events` UNIQUE idempotency.

Suspension enforcement: inside `start_agent_run` RPC gate, not UI-only; suspended users cannot queue runs.

No live trading at MVP (mandate-gated, off by default).

Fail closed: uncertain broker state, stale data, recon mismatch, risk breach → block, never guess (constitution #8).

5. THE RUN PIPELINE (Core Flow — Verified E2E)

1. User launches run (or swarm preset); `start_agent_run` RPC validates: authenticated → active subscription → BYOK key configured → rate limit (30/hr).
2. Row inserted as queued with `idempotency_key`; Realtime notifies browser.
3. Worker polls `agent_runs` WHERE `status='queued'` with FOR UPDATE SKIP LOCKED, claims one row.
4. Worker spawns subprocess: `vibe-trading run -p "<prompt>" --json --no-rich`; decrypts BYOK key server-side (`service_role`), calls DeepSeek.
5. Progress streams back (`progress_message`, `progress_iter`, `progress_at` columns).
6. Artifacts upload to Supabase Storage (`agent-artifacts` bucket, owner-scoped RLS, signed 5-min URLs).
7. Run completes/fails/timeout; status updated; Realtime notifies UI; artifacts rendered live.

Result: a run is a row, a claim, a subprocess, and a stream. That simplicity is the design's resilience.

Verification: this loop is live today. Smoke run 2026-08-22 confirmed (prompt → queued → claim → engine → completed, 24 artifacts).

6. ROADMAP — QUANT RESEARCH OS (10 Phases)

Phase	Deliverable	What it builds
0	Baseline & Freeze	Inventory, branch strategy, constitution
1	Foundation	FastAPI monolith, projects, experiments schema, research-governance substrate
2	Data Plane	Canonical schema, PIT, R2+Parquet lake, LSE OHLCV, dataset registry
3	Quant Engine	Strategy SDK, ExecutionInterface, walk-forward, backtest + emulator
4	Research AI	Experiment workflow, pre-registration UI, hypothesis labeling, promotion ladder
5	ML	Baseline models, model registry, MLflow, feature store
6	Paper Trading	Risk engine, OMS, order intents, reconciliation, paper sim
7	RL	Policy training, research-only envs, rewards, checkpoints
8	Controlled Live	Broker adapters (Alpaca/IBKR), live execution, mandate-gated
9	Scale	Prometheus/Grafana, metered billing, cost attribution
10	Expansion	Asset classes, marketplace, enterprise

Upgrade Invariant: The live run loop never breaks (D16). Every phase adds beside the existing system (strangler-fig). Each phase has a gate: design → code → test → CEO approval.

7. PHASE 1 BRIEF — FOUNDATION (Next, After MVP Launch)

7.1 Problem: Runs Have No Provenance

Today, a completed run says what happened, not why it should be trusted, on what data/params/code, or how it relates to a hypothesis. Phase 1 lays the scientific substrate: tenant-scoped experiment/hypothesis/dataset registry + provenance capture on every run, behind a FastAPI monolith, without touching the live path (D16).

7.2 In Scope

- SQL: projects, hypotheses, experiments, dataset_registry, promotion_events tables + RLS.
- Worker: provenance.py module, auto-insert experiments row on completion (gated by WORKER_PROVENANCE_ENABLED, instant rollback).
- FastAPI: platform/ at repo root, :8000, same Railway container as worker. Domains: research (registry CRUD, provenance API, health).
- Auth: Supabase JWTs (JWKS), jwt.sub = user_id, validated on every request (tenant boundary, tested like a security feature).
- Redis: Upstash (free), health + queue smoke only, fail-open (outage never blocks the loop).

7.3 Six Done-Criteria

1. Every completed run auto-writes a tenant-scoped provenance record.
2. Research API is JWT-gated (401 unauthed; cross-tenant probe returns empty/403).
3. uvicorn + worker run in ONE Railway container, both healthy.
4. Redis skeleton up; outage does NOT break the loop.
5. Run loop E2E unchanged (prompt → queued → claim → engine → artifacts → completed).
6. All test suites pass: worker 74+new · frontend build + 261/265 · engine pytest.

8. UI VISION — QUANTLAB & QUANTLAB ADMIN

Two apps, one design system. QuantLab (user) unifies Dashboard, Markets, Research, Strategies, ML Studio, RL Studio, Backtests, Paper, Portfolio, Data, Models, Live, Settings. QuantLab Admin (platform ops) unifies Overview, Users, Organizations, Billing, Infrastructure, Database, AI-Models, Data, Monitoring, Security, Audit Logs, Settings.

Design rule (locked): ONE palette, ONE typography, never lighten the dark theme. QuantLab uses dark navy + institutional blue/violet gradient + Inter/JetBrains Mono. No fake numbers on dashboard — empty states + paper-account offer when no account live yet.

9. CURRENT STATUS (SPRINT DAY 8/30, 2026-08-22)

Shipped (live on hmtrade.business):

→ MVP run loop VERIFIED end-to-end (prompt → queued → claim → engine → completed, 24 artifacts).

- ✓ Auth, Dashboard, Agent launcher, RunView, BYOK pivot (Vault), Swarm dispatch (30 presets), attachments (Premium), realtime queue, live progress streaming.
- ✓ Admin dashboard (suspend/unsuspend, audit logs, plan override).
- ✓ Paystack billing Edge Functions + E2E harness (8/9 PASS in test mode).
- ✓ Monitoring: Sentry (frontend + worker), Railway healthcheck, structured JSON logs.
- ✓ Cloudflare domain + SSL (hmtrade.business).
- ✓ Multi-provider BYOK (2026-08-22): Ollama self-hosted added alongside DeepSeek; user_api_keys supports deepseek|ollama; worker resolves provider per-user at run time.
- ✓ Landing page v2 (2026-08-22): Quant Research OS hero with trading-chart video, institutional copy.
- ✓ Phase 2 Data Plane (2026-08-22): dataset_registry + data_feeds tables, LSE adapter (httpx → Parquet → hm-datalake R2), DataCatalog UI at /data.
- ✓ Unified Account Settings (2026-08-23): Profile + Settings merged into single AccountSettings page (5 sections: Account, LLM Credentials, Billing, Notifications, Security).
- ✓ Constitution committed (SOUL, FOUNDATIONS, PROJECT, DATA,

ARCHITECTURE, WORKFLOW).

- ✓ Unified Spec adopted (docs/UNIFIED_SPEC.md, 63 sections, single source of truth).
- ✓ 10-phase Upgrade Roadmap harmonized (strangler-fig, never breaks live loop).
- ✓ PHASE1_BRIEF.md build-ready (FastAPI monolith + research-governance substrate).

Next (launch prep):

→ Paystack live-key switch (test mode → live, one-click).

🕒 Email: support@hmtrade.business mailbox, Auth templates.

🕒 Legal: ToS/Privacy with registrant block, data-protection review.

🕒 Cleanup: test accounts removed, env-vars purged on Vercel.

🕒 Replace invalid DeepSeek BYOK key (last4 ca63 returns 401 — run Profile → LLM Credentials to update).

10. BUSINESS MODEL & UNIT ECONOMICS

Revenue: flat monthly NGN subscription (₦20k/₦35k/₦75k), prepaid via Paystack.

COGS: customer's own DeepSeek spend; infrastructure serverless + one small worker (~\$100/mo MVP).

Gross margin: >90% at scale (BYOK = zero LLM cost line).

Acquisition: creator/community-led GTM (trading communities, educator channels, referral loops).

Expansion: public API + mobile views open B2B2C; Stripe enables pan-African/international.

Exit optionality: subscription SaaS with clean unit economics is acquirable by brokers, fintechs, data platforms.

The engineering is done and verified. The capital is for the company, not the code.

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